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Space: Extensive and Intensive, Actual and Virtual.

by Manuel DeLanda

There are at least two kinds of space relevant to our human identity. As biological organisms and as social agents we live our lives within spaces bounded by natural and artificial *extensive* boundaries, that is, within zones that extend in space up to a limit marked by a frontier. Whether we are talking about the frontiers of a country, a city, a neighborhood or an ecosystem, inhabiting these extensive spaces is part of what defines our social and biological identities. There are, however, other well-defined spaces which we also inhabit but which are less familiar: these are *zones of intensity*, not only those zones of temperature that define different ecosystems (from hot jungles to cold tundras), but also the zones of high pressure explored by deep-sea divers, or the zones of low gravity explored by astronauts. These other spaces are also bounded but in a different way, the limits of one zone marked by *critical points* of temperature, pressure, gravity, density, tension, connectivity, points defining abrupt transitions in the state of the creatures inhabiting those zones. Although the weather maps that have become common in television news have made intensive spaces very tangible (zones of high and low pressure, cold or warm fronts defining sharp temperature transitions) the fact remains that most philosophers have hardly thought about the questions raised by the distinction between the extensive and the intensive.

An exception is the philosophy of Gilles Deleuze, where the distinction between intensive and extensive spaces is one of the two key distinctions grounding his ontology (the other is that between actual and virtual space). The concepts come from thermodynamics where they are defined not as a distinction between spaces but between magnitudes or quantities (which can then be used to define spaces). While extensive quantities (such as volume, area, length, amount of energy or entropy) are additive, intensive quantities are not. For example, if one adds two equal volumes of water one gets twice the amount of water. But if one adds two quantities of water at forty five degrees of temperature one does not get a body of water at ninety degrees but one at the original temperature. Deleuze defines intensive quantities as “indivisible”, a definition which is simply another way of expressing the same point: a gallon of water at ninety degrees can be divided in extension, yielding, say, two half gallons, but the two parts will not each have half the temperature. {1}

Whatever way one chooses to define the terms what really matters is the reason for the lack of divisibility of intensive quantities: they are objective averages, and tend to preserve the same average value upon division. For two intensive quantities to produce a change, there must be a *difference*, or gradient, in their degree of intensity. And the change produced by a gradient of temperature (or pressure, density, speed) won't be a simple addition, but the emergence of a

spontaneous flow or movement which will tend to cancel the difference in intensity, and restore equilibrium and average values. Yet, as Deleuze emphasizes, despite the fact that it is the spontaneous cancellation of differences that explains indivisibility, the philosophical importance of intensive magnitudes can only be grasped prior to this cancellation. We can summarize what's philosophically relevant in this regard with the formula: *intensive differences are productive*. Indeed, it may be argued, wherever one finds an extensive frontier (e.g. the skin which defines the extensive boundary of our bodies) there is always a process driven by intensive differences which produced such a boundary (e.g. the embryological process which creates our bodies, driven by differences in chemical concentration, among other things.)

This approach to intensive spaces, as the site of processes which yield as products the great diversity of extensive spaces, is key to Deleuze's ontology. In what is probably his most important book, "Difference and Repetition", he writes:

"Difference is not diversity. Diversity is given, but difference is that by which the given is given...Difference is not phenomenon but the nuomenon closest to the phenomenon...Every phenomenon refers to an inequality by which it is conditioned...Everything which happens and everything which appears is correlated with orders of differences: differences of level, temperature, pressure, tension, potential, difference of intensity". {2}

It is traditional since Kant to distinguish between the world as it is given in experience to us humans, that is, the world of phenomena or appearances, and the world as it exists by itself, regardless of whether there is a human observer to interact with it. This world "in itself" is the world of "nuomena". While most philosophers do not believe in nuomena Gilles Deleuze, as the quote above illustrates, certainly does. In other words, Deleuze's ontology is a *realist* ontology. But while most realist philosophers espouse one or another form of essentialism, the belief that what gives the contents of this mind-independent world their identity is the possession of an essence, for Deleuze the identity of any being can never be taken for granted and always needs explanation in terms of the historical process which produced it. If we characterize the identity of material beings as defined by extensities (not only by its spatial boundaries but also by the amounts of matter and energy contained within those boundaries) then the process that produces those beings would be defined by intensities. In this sense, human beings not only inhabit extensive spaces, they themselves are extensive spaces. Generalizing this to include mental phenomena would involve defining psychological intensities (not only grief, joy, love, hate, but also beliefs and desires which also come in different intensities) as well as the corresponding extensities. In this essay I will avoid this important issue and stick to our bodily identities which do form extensive spaces in a straight forward sense.

Despite the fact that Deleuze takes the distinction between the extensive and the intensive from physics, in the page following the quote above he argues that

nineteenth century thermodynamics cannot provide the foundation he needs for his ontology. Why? Because that branch of physics became obsessed with the final equilibrium state (defined by its amount of entropy, an extensive quantity) at the expense of the difference-driven intensive process which gives rise to that state. Fortunately, this shortcoming of classical thermodynamics has been fixed today in the latest version of this branch of physics, appropriately labeled “far-from-equilibrium thermodynamics”, and the effect of this repair has been to make this discipline all the more interesting philosophically. In a nut shell, while equilibrium thermodynamics focuses on what happens once the intensive differences have been canceled, far-from-equilibrium thermodynamics studies systems that are continuously traversed by a strong flow of energy or matter, a flow which does not allow the differences in intensity to be canceled, that is, a flow that maintains these differences and keeps them from canceling themselves out. In a sense, the new field studies systems in a *zone of higher intensity*, and it is only in this zone that difference-driven morphogenesis comes into its own, and that matter becomes an active agent, one which does not need form to come and impose itself from the outside, as is the case with essentialism. In short, only in this zone of intensity can we witness the birth of extensity and its identity-defining frontiers.

I said above that the distinction between extensive and intensive spaces is one of two distinctions which are fundamental in Deleuze’s realist ontology. Far-from-equilibrium thermodynamics also throws light on the second distinction, that between *actual and virtual* space. Even in equilibrium thermodynamics scientists must face the fact that, given a system in which an intensive difference exists, the final state of equilibrium to which the system tends is somehow already “present” prior to its actualization. That is, the final state acts as an “attractor” for the process and explains the tendency of the intensive difference to cancel itself. But what ontological status does that final state have prior to its coming into actual existence? One may think that the category of “the possible” is the ontological category one would need to describe this status, but this would be wrong. Despite what some essentialist modal logicians may argue, possibilities are not mind-independent entities, though they do possess psychological reality (no one can deny that human beings can in fact entertain various possible scenarios in their heads). States acting as “attractors”, on the other hand, possess a certain objective efficacy even while not being fully actual, since they guide real processes towards a definite outcome prior to the latter’s actualization.

One way of approaching the ontological status of attractors would be to say that while they are not possibilities they do act as *the structure of a space of possibilities*. Of all the possible outcomes only one, or a few, become regularly actualized, a fact that suggests that the space of possible outcomes is greatly constrained, or in other words, that it has structure. While the possibilities making up this space are not real (other than in a purely psychological way) the structure of the space may be considered fully real and mind-independent. But if this reality is not actual (by definition) what is it? Deleuze’s answer would be that it is virtual, not in the sense of a virtual reality (as exemplified by computer simulations, or

even cinema) but in the sense of a *real virtuality*. In this regard, the contribution of the new thermodynamics is that only in the zone of intensity it explores do physical processes display the full repertoire of attractors. While in linear systems near equilibrium only steady-state attractors exist, nonlinear far-from-equilibrium systems display steady-state, periodic and chaotic attractors. Moreover, instead of the single global equilibrium of the classical theory we now have multiple equilibria, which means that history matters. While with a single possible outcome the different paths followed by systems on their way there can be ignored, with multiple possible outcomes the details of the history followed do matter (this is what physicist call “path dependence”). Why is all this important? Because once this rich structure is revealed it becomes harder for philosophers to ignore the ontological questions raised by it.

To tackle this ontological issue, however, we need to go beyond physics and into mathematics to define the status of virtual space. The mathematical distinction that we need is that between *metric and non metric* spaces, that is, spaces in which the concept of “length” is fundamental and spaces in which it is not. Mathematically a space is defined by a set of points and a definition of “relations of proximity” between points, in other words, of the relations which define a given subset of the points as a neighborhood. If proximity is defined via a minimum length (e.g. all points less than a given distance away from a center form a neighborhood) the space is said to be metric (whether flat, as in Euclidean geometry, or curved, as in the non Euclidean versions). If some other criterion is used the space is said to be non metric (as in projective, differential or topological geometries). What other criterion of proximity could be used? In differential geometry, for example, one takes advantage of the fact that the calculus operates on equations expressing rates of change and that one of its operators (differentiation) gives as its output an instantaneous value for that rate of change. The points that form a space can then be defined not by rigid lengths from a fixed coordinate system (as in the metric case) but by the instantaneous rate at which *curvature* changes at that point. Some parts of the space will not be changing at all, other parts changing slowly, while others changing fast. A differential space, in effect, becomes a *field of rapidities and slownesses*, and via these infinitesimal relations one can specify neighborhoods without having to use rigid lengths. Mathematicians refer to such a differential space as a “manifold” or a “multiplicity”.

To Deleuze this notion of “multiplicity” has several important features. First of all, when Gauss and Riemann introduced the notion in the first half of the nineteenth century, they revolutionized the very way in which we can pose spatial problems. (Einstein and others would several decades later use these new problem-posing resources to, in turn, revolutionize our ideas of physical space-time.) One feature of this revolution was to get rid of the idea that a space of a given number of dimensions (say, a two dimensional folded sheet) must be inscribed within a space one dimension higher (a three dimensional box) in order for it to be properly studied. The need for the extra dimension arises from the procedure of assigning

Cartesian coordinates to every point of the sheet (via rigid lengths expressing the distance of each point to one of the three coordinates). But if the sheet can be studied using only local information (the rapidity or slowness at which curvature is changing at a given point) the need for such global embedding space is eliminated. This, Deleuze argues, also eliminates the extra dimension which relates to the space being studied as a transcendental dimension. As he puts it, "In all cases the multiplicity is intrinsically defined, without external reference or recourse to a uniform space in which it would be submerged". {3}

Eliminating a global embedding space and viewing all spaces in purely local terms is crucial for Deleuze because to him this is not a formal issue in the philosophy of mathematics but an ontological issue bearing directly on the status of the virtual. The virtual structure of possibility spaces must never be made into something transcendent but always conceived as *immanent* to the material world. A multiplicity, as writes, "however many dimensions it may have, never has a supplementary dimension to that which transpires upon it. This alone makes it natural and immanent." {4} But how, one may ask, can such a geometric object have ontological significance? In what sense can it act as the structure of a space of possibilities? The answer is that multiplicities or manifolds can be used to study a physical system if each of their dimensions are assigned values from one of the "degrees of freedom" (or relevant ways of changing) of the system itself. That is, the multiplicity or manifold becomes the space of all the possible states which a given system can have. (This space of possible states is referred to as "state space" or "phase space"). {5} And more importantly, while the points in the manifold represent all the possibilities for a given system, certain topological features of this space represent the invariant structure of that space. These *topological invariants* are what I referred to above as "attractors". While state spaces themselves are nothing but mathematical representations (and the possibilities they symbolize are not mind-independent realities) their topological invariants (their dimensions, their singularities or attractors) may indeed be interpreted as being fully real.

How one goes from a mathematical representation (a manifold) to a real non-actual entity (a virtual multiplicity) is a complex issue which I cannot tackle in the space of this essay, but which I have addressed in detail elsewhere. {6} But even if it was clear just how such an ontological move can be made that would take us only half way through an account of virtual space, the space formed by all multiplicities (the plane of consistency or plane of immanence). A full account needs to introduce yet another unfamiliar virtual entity (referred to by various names such as "abstract machine", "line of flight", "quasi-causal operator") which job is to continuously extract multiplicities from the systems in which they are actualized and mesh them together without reducing their heterogeneity (that is, give them consistency as a space). {7} Again, just how these two operations are supposed to be performed cannot be discussed here but they are crucial if virtual space is to be truly immanent and not an eternal reservoir of fixed archetypes. In other words, philosophers cannot simply use the label "immanent" and assume this

is going to eliminate transcendence, they must give concrete mechanisms of immanence. {8}

Let's assume for a moment that one can give a full account of what virtual space is and how it is constantly being produced and reproduced. What materialist world view emerges from these ideas? In a nut shell, the world would consist of a topological (non metric) space containing all the constraints which organize the physical, chemical, biological and social processes which produce the actual systems (planets, molecules, species, institutions) which inhabit our familiar Euclidean metric and extensive space. Non metric and metric spaces would be connected via intermediate spaces which would be mostly intensive. Thus, the virtual, the intensive and the actual would constitute the three spheres of reality, with virtual multiplicities constraining and guiding intensive processes which in turn yield specific actual entities. The opposite movement, from the actual to the intensive to the virtual, would also be constantly happening, guaranteeing the independence and immanence of multiplicities. Within this material world human thinkers of different types would follow one or the other movement, scientists tracking the actualization of the virtual (and focusing on actual beings, as well as intensive processes) while philosophers would track the opposite motion, that which reconstitutes virtual multiplicities (as ideal events) out of actual entities and gives them consistency as a space. To end this essay with Deleuze's own words:

“It could be said that science and philosophy take opposed paths, because philosophical concepts have events for consistency whereas scientific functions have states of affairs or mixtures for reference: through concepts, philosophy continually extracts a consistent event from the states of affairs....whereas through functions, science continually actualizes the event in a state of affairs, thing, or body that can be referred to.” {9}

References:

{1} Actually Deleuze defines the intensive not as “indivisible” but as “what cannot be divided without changing nature”, a definition that acknowledges the fact that critical points do subdivide an intensive line of values but only by marking the onset of an abrupt change of state. As he writes:

“What is the significance of these indivisible distances that are ceaselessly transformed and cannot be divided or transformed without their elements changing in nature each time? Is it not the intensive character of this type of multiplicity’s elements and the relations between them? Exactly like a speed or a temperature, which is not composed of other speeds or temperatures, but rather is enveloped in or envelops others, each of which marks a change in nature. The metrical principle of these multiplicities is not to be found in a homogeneous milieu but resides elsewhere, in forces at work within them, in physical phenomena inhabiting them...”

Gilles Deleuze and Felix Guattari. *A Thousand Plateaus*. (University of Minnesota Press, Minneapolis, 1987). Pages 31-33.

This quote uses the term “distance” as if it was a non metric property, though in its usual meaning it certainly denotes something metric. Deleuze takes this special intensive meaning of “distance” from Bertrand Russell.

{2} Gilles Deleuze. *Difference and Repetition*. (Columbia University Press, New York 1994). Page 222.

{3} Ibid. Page 183.

Elsewhere he writes:

“Unity always operates in an empty dimension supplementary to that of the system considered (overcoding)...[But a] multiplicity never allows itself to be overcoded, never has available a supplementary dimension over and above its number of lines, that is, over and above the multiplicity of numbers attached to those lines.”

Gilles Deleuze and Felix Guattari. *A Thousand Plateaus*. Pages 8-9.

{4} Gilles Deleuze and Felix Guattari. Ibid. Page 266.

This remark is made about the "plane of consistency" not about multiplicities. But the former is nothing but the virtual space formed by the multiplicities themselves.

{5} When Deleuze defines his multiplicities he always seems to be referring to manifolds whose dimensions are used to represent degrees of freedom (or independent variables) of some dynamic, and not to manifolds as mere geometric objects. Thus, in his first introduction of the term he says:

"Riemann defined as 'multiplicities' those things that could be determined by their dimensions or their independent variables. He distinguished between discrete multiplicities and continuous multiplicities. The former contain the principle of their own metrics.... The latter found a metrical principle in something else, even if only in phenomena unfolding in them or in the forces acting in them."

Gilles Deleuze. *Bergsonism*. (Zone Books, New York, 1988). Page 39.

And elsewhere he says, using the word "Idea" to refer to concrete universals or virtual multiplicities as replacements for essences:

"An Idea is an n-dimensional, continuous, defined multiplicity. Colour -or rather, the Idea of colour - is a three dimensional multiplicity. By dimensions, we mean the variables or co-ordinates upon which a phenomenon depends; by continuity, we mean the set of relations between changes in these variables....by definition, we mean the elements reciprocally determined by these relations, elements which cannot change unless the multiplicity changes its order and its metric."

Gilles Deleuze. *Difference and Repetition*. Op. Cit. Page 182.

{6} Manuel DeLanda. *Intensive Science and Virtual Philosophy*. (Continuum Press, London, 2002). Pages 30-38.

{7} Gilles Deleuze and Felix Guattari. *A Thousand Plateaus*. Page 72.

"There was a first group of notions: the Body without Organs or destratified Plane of Consistency; the Matter of the Plane, that which occurs in the body or plane (singular, nonsegmented multiplicities composed of intensive continuums, emissions of particle-signs, conjunctions of flows); and the Abstract Machine, or Abstract Machines, in so far as they construct that body or draw the plane or 'diagram' what occurs (lines of flight, or absolute deterritorialization)."

{8} The term "mechanisms of immanence" does not, to my knowledge, occur in Deleuze's work, but he expresses himself in similar ways:

“Many movements, *with a fragile and delicate mechanism*, intersect: that by means of which bodies, states of affairs, and mixtures, considered in their depth, succeed or fail in the production of ideal surfaces [plane of consistency]; and conversely, that by means of which the events of the surface are actualized in the present of bodies (in accordance with complex rules) by imprisoning their singularities within the limits of worlds, individuals and persons.”

Gilles Deleuze. *Logic of Sense*. (Columbia University Press, New York, 1990). Page 167. (My italics).

{9} Gilles Deleuze and Felix Guattari. *What is Philosophy?*. (Columbia University Press, New York, 1994).Page 126.